Response to Reviewer 1 for "Snow accumulation variability limits InSAR SWE retrieval"

Zachary Hoppinen[1], Simon Zwieback[1], Ross Palomaki[2], and Hans-Peter Marshall[3]

[1]Geophysical Institute, University of Alaska Fairbanks, 2156 Koyukuk Drive, Fairbanks, AK
[2]Institute of Arctic and Alpine Research, University of Colorado, 4001 Discovery Dr, Boulder, CO 80303, USA
[3]Department of Geosciences, Boise State University, 1910 University Drive, Boise, ID 83725, USA

Correspondence: Zachary Hoppinen (zmhoppinen@alaska.edu)

This is a very interesting and timely analysis of the limitations of InSAR SWE retrieval. The authors provide great motivation for this study and tie it to snow applications relatable to Cryosphere readers. It is unique in its focus on specific InSAR processing effects over complex terrain where the underlying snow depth and water content distributions are highly variable. The validation study with aerial lidar is appropriate and well thought out. My main overall comment is that this is very heavy in remote sensing theory and the general Cryosphere audience could use more context and explanation, especially as this relates to NISAR. The Theoretical Framework section assumes a lot of familiarity with SAR and InSAR processing at a level that some, used to polished science data products, may lack. I think much of this could be resolved with the inclusion of one or more figures that help explain the windows, multi-look windows, averaged windows, and sub-windows, and how these relate to the "pixels" we'd see in a NISAR data product. The Discussion section does a great job capturing the major points and takeaways for other researchers.

Thank you for taking the time to read and give this thoughtful review of our paper. Find each point addressed below. Line numbers refer to the revised draft.

Specific Comments

Theory (2): I would like a graphical representation that clearly explains the windows, multi-look windows, averaged windows, and sub-windows. The concept of windows is brought up with little context, and at first I was confusing it between spatial and temporal windowing. For the science user base interacting with NISAR data, it would be nice also graphically explain the relation between these window definitions and pixels. If multi-look averaging is a step in NISAR's data processing I'd like to see a more detailed description of it. I'm not sure if it is using adjacent frames from along NISAR's orbit or a pair of SAR collections. Is Figure 1 showing sub-windows that end up as a single pixel in an InSAR data product?

Thank you, we appreciate this is a central idea and worth the space to explain clearly. We have added an example figure (Figure 1) showing the spatial averaging and window sizes using a zoom from the December pair of NISAR acquisitions. Section 2 now opens with a paragraph that defines the resolution cell, the multi-look window, and the product pixel, and states the two window sizes NISAR uses for its 20 m and 80 m layers (lines 75–81). Yes, each panel of the phasor figure (now Figure 2) is one multi-look window that becomes one pixel of the product, and its caption now says so.

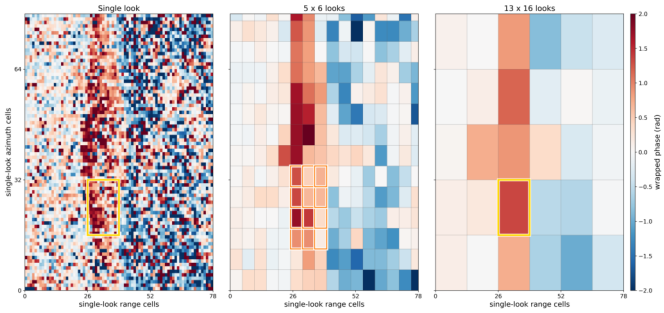


Figure 1. Added as suggested to explain the window concept (Figure 1 of the revised manuscript).

Theory (2): I would consider not setting ϕ=$\Delta\phi$ since it is used so few times, and Section 3.4 seems to revert to the $\Delta\phi$ notation.

We agree and have removed the $\phi = \Delta\phi$ shorthand. Section 2, including Equations 2 and 3 and the Figure 2 caption, now uses $\Delta\phi$ throughout, matching the Introduction and Section 3.4 (lines 90–99).

Methods (3.1, 3.2): Can we get more information on the LiDAR system used, i.e. the platform and sensor, flight height, GNSS positioning? Also, define how snow depth and SWE are calculated. I assume snow depth comes from digital terrain differencing, but what algorithms are used? And why do snow depth and SWE have different resolutions (3m and 50m)?

Yes, here are some details from the two papers we could find and an email response from ASO. We have added citations describing the ASO lidar system and products to Section 3.2 (lines 158–161) and expand with more detail here, based on Painter et al. (2016) and updated details provided by ASO (J. Deems, personal communication, 2026). The flights used here (2023–2026) were collected with a Riegl VQ-1560 II-S airborne laser scanner, positioned with a lidar-integrated Trimble Applanix AP60 GNSS/IMU differentially corrected with Applanix RTX. Each survey is flown at a constant altitude, generally 5800–7300 m (19,000–24,000 ft) above sea level, so the height above ground and the ground point density varies with the terrain.

Snow depth is computed by subtracting gridded snow-free surfaces from gridded snow-on surfaces. Following Section 5.2.2 of Painter et al. (2016), separate surface pairs are built from first returns in open terrain and from ground points classified under forest canopy, and the two resulting depth grids are merged using vegetation masks. SWE is the product of snow depth and a bulk snow density simulated by the iSnobal model, which assimilates the observed snow depth map and adjusts based on in-situ density observations.

ASO grids snow depth at 3 m from the lidar data, while SWE is delivered at 50 m, the resolution of the iSnobal model that simulates the snow density. We use the finer 3 m depth to keep the most spatial detail and convert it to ΔSWE with a density taken from the 50 m products (Section 3.2).

We don't include the emailed information in the main text as The Cryosphere discourages personal communication citations.

Methods (3.3): Provide a graphical representation of the radial power spectral density for ΔSWE spatial variation.

This figure already exists as Figure 5 (Figure 3 of the original submission), which shows the radial power spectral density of the ΔSWE field for every 1.15 km window with the fitted two-segment power law and the break wavelength. It was placed with the results, but Section 3.3 did not point to it. Section 3.3 now refers to it where the radial spectrum is introduced (line 170).

Methods (3.3): How is 95% snow cover determined? Is this what Appendix B covers? If so, specify.

The 95% criterion comes from the lidar snow mask, not from the VIIRS snow cover of Appendix B. A window is kept for the spectral analysis when at least 95% of its cells pass the lidar snow test of Section 3.2 in all four pairs. Section 3.3 now states this (line 179). Appendix B uses VIIRS only as an independent check on the snow coverage from the NISAR coherence residual.

Methods (3.4): How is κ calculated?

We use the dry-snow approximation of Leinss et al. (2015), $\kappa = (2\pi/\lambda)(1.59 + \theta^{5/2})$ with θ in radians, which gives 52.7, 224.1 and 15.5 rad m^{-1} of ΔSWE at L-, C- and P-band for the 40° incidence angle used throughout. Section 3.4 now states the formula as Eq. (9) and the wavelength and incidence angle it is evaluated at (lines 200–202). Leinss et al. (2015) include a correction factor α that they set to 1. We omit it rather than introduce a symbol only to remove it.

Methods (3.5): arg(M) and κ have been described, but the introduction of arg(M)/κ could use more description.

$\arg(M)$ is a phase, and dividing by κ turns it into an SWE error. Section 2.1 now defines $|M|$ as the coherence factor and $\arg(M)$ as the retrieval bias, the offset of the averaged phase from the phase of the window-mean ΔSWE (lines 126–128), and Section 3.5 refers back to it where $\arg(M)/\kappa$ is reported.

Methods (3.6): Again, a figure showing how multi-looking works specifically for NISAR would be helpful for potential NISAR users.

The new Figure 1 shows this for NISAR directly. Section 3.6 now refers to it where the 5×6 and 13×16 look counts are introduced (line 250). The single-look interferogram in that figure was formed from the delivered NISAR RSLCs with the same ISCE3 workflow that produces the standard GUNW (see the related repository for the code), but with the multi-looking step turned off, so the three panels differ only in the window size.

Methods (3.6): In Equation 9, what does the ˜ over γ mean for true coherence? One line 199 true coherence was called simply γ.

The tilde marks the bias-corrected estimate of the true coherence γ, obtained by inverting the Bamler and Hartl (1998) expectation, and the hat the sample coherence delivered in the product. Section 3.6 now defines all three where Eq. (12) is introduced (lines 264–266), and the corrected difference is written $\tilde{\gamma}_{20} - \tilde{\gamma}_{80}$ throughout.

Results: Figure 4, indicate where there is higher elevation / more complex terrain, perhaps in a separate figure of the basin.

We have added a study area figure (new Figure 3) in Section 3.1 showing elevation with hillshade over the lidar coverage, along with the CDEC snow pillows, the water pixels used for the coherence calibration, and the NISAR acquisition geometry. The Results now point to it where the terrain pattern of $|M|$ and $\arg(M)$ is described (lines 291–293).

Discussion: To what extent does the NISAR data being uncalibrated affect this study?

Very little, because the study uses only the NISAR coherence magnitude and never retrieves phase or SWE from the NISAR products. Since submission, NISAR has released provisional products for both pairs (product version 1.0.11, with RFI mitigation and updated geocoding), and we have redone the NISAR analysis with them. Over the lidar extent the provisional coherence is slightly lower than the beta coherence, by a mean of 0.024 for the December 7–19 pair and 0.009 for the December 19–31 pair, and within each pair the change is the same at the 20 and 80 m look counts (within 0.001). Recalibrating the effective number of looks over water on the provisional products moves the oversampling factor from 1.29 to 1.32. The corrected $\tilde{\gamma}_{20} - \tilde{\gamma}_{80}$ residual is unchanged for the no-accumulation pair (median -0.004) and changes from $+0.031$ to $+0.028$ for the storm pair. The revised manuscript uses the provisional products throughout, and the calibration values, the coherence results, the coherence figure, and Appendix B have been updated (lines 253–258 and 326–327).

Technical Corrections

Line 32: No comma needed after "acquisitions".

Removed (line 32).

Line 36, 45: Consider alternative to having "SWE" as single variable name.

We considered this but have kept ΔSWE. It is the standard notation in the snow InSAR literature (Guneriussen et al., 2001; Leinss et al., 2015; Oveisgharan et al., 2024). Also, a single-letter symbol would be less immediately clear to the reader. We set it upright, as SWE, so that it reads as one quantity rather than a product of variables.

Line 125: Specify that 3m and 50m are spatial resolutions, if that is what they are.

They are. Section 3.2 now says 3 m and 50 m spatial resolution where the two products are introduced (line 158).

Response to Reviewer 2 for "Snow accumulation variability limits InSAR SWE retrieval"

Zachary Hoppinen[1], Simon Zwieback[1], Ross Palomaki[2], and Hans-Peter Marshall[3]

[1]Geophysical Institute, University of Alaska Fairbanks, 2156 Koyukuk Drive, Fairbanks, AK
[2]Institute of Arctic and Alpine Research, University of Colorado, 4001 Discovery Dr, Boulder, CO 80303, USA
[3]Department of Geosciences, Boise State University, 1910 University Drive, Boise, ID 83725, USA

Correspondence: Zachary Hoppinen (zmhoppinen@alaska.edu)

General Comments:

Thanks to the authors for this contribution. The idea presented is relatively new to the dInSAR technique for dSWE and it presents interesting discussion on multilooking window selection for processing. However, I have found issues that prevent me from recommending its publication without a major revision. The paper needs to broaden theoretical discussion, including selected references and provide a more thoughtful discussion on their findings.

Thank you for taking time to read and give this thoughtful review. We agree with almost all of your points and think the text is much clearer for your comments. All line, figure, and section numbering refer to the revised version.

There are important papers that should be consulted as they discuss coherence for nonstationary processes. Touzi et al., 1999 models differently the non-stationarity. Yueh et al., 2026 directly addresses the topic presented here. The latter is particularly important here. Also consider Rott et al., 2003, although the formulation is for sub-pixel scale.

Thank you. All three are now cited, together with Zebker and Chen (2005), which treats the same estimation problem: Rott et al. (2003) and Yueh et al. (2026) in the introduction and Section 2.1 as the Gaussian analytic treatments whose result is the first term of our expansion, Touzi et al. (1999) and Zebker and Chen (2005) in Sections 2 and 3.6 for the stationarity requirement and the two competing biases (lines 60, 86, 106, 134, and 238).

An important addition to the paper would be to quantify both the absolute value and the argument of M with respect to standard deviation or relevant metrics within the window. Now, the reader cannot quantify the any impact of the variability. It is necessary to better quantify the magnitude of the effect. Consider the following:

- *Choose a window size and generate phase samples ("per-cell deviation from mean")*
- *Calculate complex M, repeat enough times till convergence.*
- *Repeat for different values of sigma. This would provide an idea of the effect of M for different window variabilities, instead of the arbitrary examples presented in Figure 1.*

This can also be calculated analytically but of course it's more tedious. Consider this both in absolute dSWE and radians. Consider different distributions. For example, in case the underlying distribution is Gaussian of δ or a symmetric distribution

in general, I would expect that Earg(M) = 0. This is your case in Figure 1 (middle). Then, the estimated phase from the window will not be skewed. The reader needs to understand already in section 2 what causes the skewing.

How does the typical multilooking window look like in terms of ΔSWE Which distribution approximate it better? This could be linked to your dataset. Based on your results (Figure 5), the distribution of ΔSWE may not be symmetrical. While this is partially addressed in Section 4.2, I think the aggregation over the whole dataset does not allow understanding of the individual multilook variability.

We agree that the reader needs a number. Section 2.1 now expands $\log M$ in the cumulants of the within-window phase distribution, giving $|M|$ and $\arg(M)$ to leading order (Eqs. 7 and 8), and Figure 7 in Section 4.2 plots both against the phase spread over all lidar windows. The first term is the Gaussian result of Rott et al. (2003) and Yueh et al. (2026). The second is the bias, which a Gaussian cannot produce and which the odd moments cause. We do use the lidar windows as the background of Figure 7 in addition to your theoretical suggestion, as we feel that grounding the results in the actual observed ΔSWE values is useful. Figure 7 shows every window individually rather than an aggregate. We have not fitted a named distribution, because beyond about 1.5 rad of spread the phase wraps and M depends on the full shape of the distribution rather than on any low-order family.

Another important consideration are variations of local incidence angle in the multilooking window. Although the sensitivity is smaller than ΔSWE changes in LIA within the multilooking window will produce the same effect even for constant ΔSWE In particular to your dataset, the northeast area seems to have lower values of M, but also presents more topographical variations.

We use a constant local incidence angle of 40° (Section 3.4) over the lidar extent for the ASO-based conversions, so this is not a factor in our results, and the low $|M|$ in the northeast of the basin is ΔSWE structure rather than terrain geometry. We have added a sentence to Section 3.4 to clarify this and a sentence to the Limitations about the potential for local incidence angle variations to add to or reduce the within-window effects over real terrain. The NISAR analysis, since it compares the storm pair against the no-accumulation pair, holds the local incidence angle constant in that part of the comparison, though the larger spread of incidence angle within an 80 m window than within a 20 m window might affect it.

Can you report the spatial baseline between NISAR acquisitions? The orbital tube for NISAR is small, but I wonder if small spatial baseline between passes can impact the coherence in some areas of your images.

The perpendicular baselines, read from the GUNW metadata over the lidar extent, are -109 m (range -111 to -106 m) for December 7–19 and 127 m (123 to 131 m) for December 19–31, with parallel baselines of -69 and 105 m. At the basin's 44.7° incidence angle and 1009 km slant range these give heights of ambiguity of 790 m (700 to 880 m) and 680 m (600 to 760 m). The critical baseline for the 40 MHz range bandwidth is 27 km at the 40° local incidence angle used in the analysis (32 km at the basin's 44.7° metadata incidence on flat terrain), so the geometric (spatial) decorrelation is 0.004 to 0.005 for both pairs. For a slope facing the radar at a 10° local incidence angle it rises to about 0.02. Spatial decorrelation is therefore negligible. Section 3.1 now states the baselines and heights of ambiguity (line 151).

A distinction between posting, resolution and multilook is needed. The multilooked domain, either 5x6 or 13x16, does not correspond to 20 or 81 ground meters, as multilooking is calculated in radar coordinates. Range resolution also depends with incidence angle. Comparing NISAR coherence and Lidar-derived M needs to be done with care (e.g., Section 4.3).

We agree, and we have made the distinction explicit and standardized the wording. Section 3.4 now states this distinction where the lidar window sizes are chosen, and says that the square 81 and 21 m lidar windows are close proxies for the two NISAR windows and that the analysis addresses the multi-look window on the snow side rather than reproducing the NISAR geocoding (lines 212–215). Throughout the manuscript we now use "window" and "window size" for the averaging block, "posting" for the 20 and 80 m GUNW grids and "resolution" for the single-look cell and the ASO products, and have removed "footprint" and "multi-resolution", including from the section titles of Sections 3.6 and 4.4. We think this change was a great suggestion that really clarifies the text especially combined with the new Figure 1 suggested by reviewer 1.

The terminology "within-window", "sub-window", and "footprint" is used somewhat interchangeably throughout the manuscript.

We agree this was confusing for a reader. See above comment where we also remove footprint. We have also standardized to within-window and removed sub-window throughout.

Specific Comments:

The abstract and introduction does not stress enough the main problem. Phase is highly sensitive to ΔSWE and ΔSWEis highly spatially variable. Also high decorrelation makes large multilooking an attractive option to deal with noise at a cost of spatial resolution.

We agree. The introduction now states the phase sensitivity after Eq. (1) and opens the deformation-gradient paragraph with why retrievals multi-look and what that assumes (lines 37–38 and 46–52). The spatial variability of ΔSWE was already stated (lines 39–40).

Section 2: Somewhere along the first paragraph it should be commented that this "averaging" is performed spatially.

We updated the text in the first paragraph and added figures to clarify what is meant by averaging, and that it is performed spatially (lines 46–52 and Figure 1).

Equation 2 should be introduced more clearly.

The lead-in to Eq. (2) now says why the phase is decomposed and defines δ as the difference from the window mean (lines 90–94). The $\phi = \Delta\phi$ shorthand has been removed and $\Delta\phi$ is used throughout.

Figure 1 requires more explanation, describe how many pixels are you using, the central frequency you are using (as later described this is critical). Also consider discussing if the presented dynamic range is reasonable for SAR revisit times. Do you observe $\pm$8cm=80mm ΔSWE often?

This is an illustration of the effects we discuss later. The three panels and their values are chosen to isolate the constant, symmetric, and skewed cases synthetically, not observations, and we have kept the figure free of data detail for that reason. The caption now states that each window holds 12×12 cells and that the phasors use the L-band κ of Section 3.4 (52.7 rad m^{-1} at 40°). The $\pm$8 cm range is large but realistic for these pairs: the within-window standard deviation of ΔSWE at 81 m has a median of 3.4 cm and a 1st to 99th percentile range of 0.9 to 23 cm, so cells differing by 8 cm within one window are possible, and the basin-median ΔSWE of the largest event is 23 cm. The figure is also clearer to the reader for having a larger spread.

I think you need to differentiate between real noise free ΔSWE or $\Delta\phi$ and observed values. For example, in Line 86 it is unclear if δ refers to stochastic noise or ΔSWE deviations from the mean. If the process is stationary, then δ = 0 for every

pixel. The section already discusses in terms of estimated phase variance when in my understanding, you are discussing the effect of the underlying distribution of ΔSWE

Agreed, δ was used for two things. The paragraph after Eq. (4) is rewritten so that Eqs. (2) to (4) are noise-free, δ is the ΔSWE deviation only, and decorrelation noise stays in γ_{other}. The lead-ins to Eqs. (3) and (6) no longer say "observed" (lines 102–103, with the lead-ins at lines 95 and 122).

Section 3.4: When using the LIA=40 for the whole image, the topography is ignored, which varies subwindow variability.

Agreed. As noted in our reply to the general comment above, the constant 40° incidence angle means the reported $|M|$ and $\arg(M)$ isolate the ΔSWE variability alone, which Section 3.4 now states (line 205). The new Limitations paragraph quantifies what within-window variation of the local incidence angle over real terrain would add and notes that the within-window loss over steep terrain may be somewhat smaller or larger than reported (lines 517–522).

Section 3.6: The claim on the Circular Gaussian as the limitation is unclear.

The sentence now says what the assumption is, fully developed speckle from many independent scatterers per cell, and that cells with only a few strong scatterers depart from it, where the bias correction is only approximate (lines 249–250).

Equation 8 is based on stationary samples. Since the central discussion of the paper is the non-stationarity I think commenting the validity of this equation is in order.

Agreed that Eq. (11) assumes looks that share one coherence and one phase. We apply it as if each window had the effective coherence $|M|\gamma_{\mathrm{other}}$, with M the mean phasor of that window's own cells. We checked this by simulation (stats/plot_bamler_nonstationary.
in the repository), drawing looks from the same speckle model Eq. (11) assumes but giving each window its own within-window phase pattern. Figure 1 shows that the de-biased sample coherence tracks the window's true $|M|\gamma_{\mathrm{other}}$ closely at both look counts, across the range of coherence and spread relevant here, while the raw sample coherence sits above it. We have kept this in the reply rather than the main text, where Section 3.6 is already dense, but would add a sentence there if the referee prefers.

Section 4.5: The results, in particular related to C-band are quite pessimistic. How do these compare to literature, for example Oveisgharan et al., 2024, Oveisgharan et al., 2026, Bovenga et al., 2025 among others? This requires reconciliation with previous papers.

The C-band result is pessimistic for large events, and we think that is likely an accurate estimate of C-band ability to resolve these large events. Eq. (8) shows this: C-band keeps a coherence factor above 0.3 only where the within-window ΔSWE spread is below about 0.7 cm, whereas the Tuolumne storm windows have a median spread of 3.4 cm. The C-band retrievals of Oveisgharan et al. (2024, 2026) and Bovenga et al. (2025) were made in snow climates with smaller accumulations, where that condition might be met, and our Limitations describe the Tuolumne storm pairs as an upper bound. For major accumulation events in complex terrain, our result implies that within-window variability would probably prevent a C-band retrieval, which calls for testing with lidar pairs in the basins where C-band has been applied. Section 5.4 now states this and lists that test as important future work (lines 455–458).

Section 5.2: I don't agree with this. For tower imagery I would not expect high spatial variability within the multilook window. The whole window may be smaller than a typical satellite resolution cell.

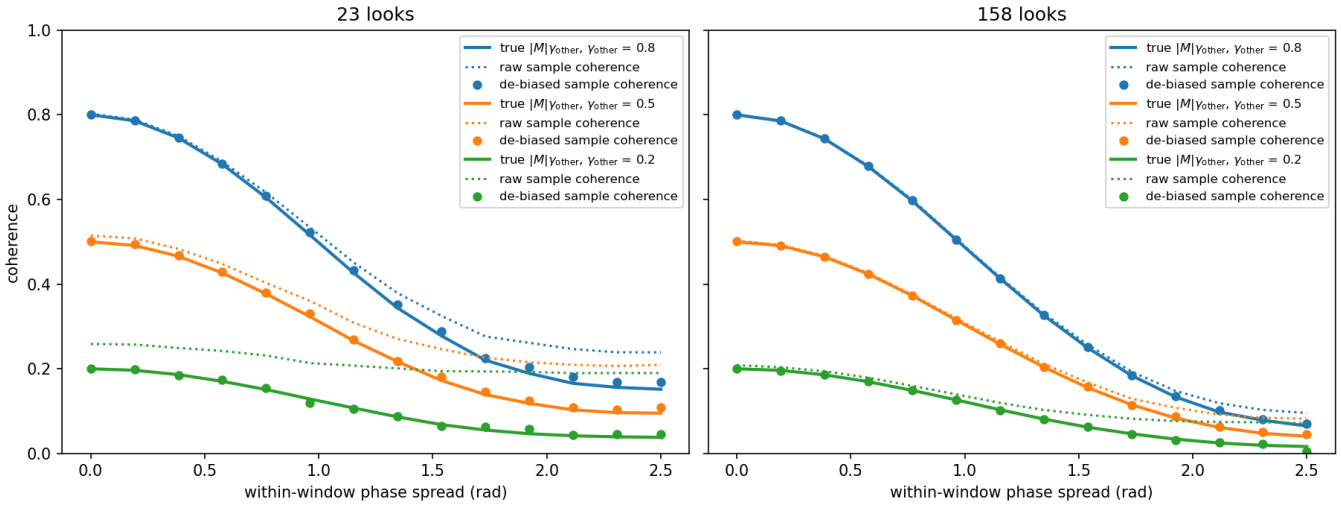


Figure 1. True effective coherence of each window, $|M|\gamma_{\mathrm{other}}$ with M the mean phasor of its own cells (solid), raw mean sample coherence (dotted), and Bamler-de-biased sample coherence (markers) against the within-window phase spread, for 23 looks (**left**) and 158 looks (**right**) and γ_{other} of 0.8, 0.5, and 0.2. Monte Carlo over 3000 windows per point, each with its own Gaussian phase pattern.

Agreed. SodSAR images 0.1 m pixels over a scene of about 35 by 30 m (Ruiz et al., 2020), below a single satellite resolution cell, so its wind dependence is not evidence for variation between cells across a multi-look window. The paragraph no longer claims that, and now places the tower result at the sub-cell scale of Rott et al. (2003) (lines 417–418).

Snow depth/SWE for sea ice snow, is typically smaller than land. This is hinted on line 433. This is why I believe you need to quantify either analytically or numerically the effect of phase variance in the complex coherence.

Agreed on both points. The quantification is now Eq. (8) and Figure 7: the coherence loss is set by the within-window phase spread, $|M| \approx e^{-\kappa^2\sigma^2_{\Delta\mathrm{SWE}}/2}$, and the bias by its skew. The smaller snow depth and ΔSWE over sea ice enter directly through $\sigma_{\Delta\mathrm{SWE}}$, so for the same wavelength and window a sea-ice snowpack with a few centimeters of accumulation and a small within-window spread sits far up the curve of Figure 7 where the Gaussian term holds and the bias is negligible, whereas the Tuolumne storm windows, with a median spread of 3.4 cm, sit past the wrap limit at C-band. Our reference to Ming et al. (2026) is to the mechanism they identify, not to the magnitude, which Eq. (8) shows must be smaller over sea ice.

Specific Comments:

Line 22: how "approximately weekly observations" are meet? Which constellation? S1? Where?

The weekly figure refers to NISAR. Each track repeats every 12 days, but most locations lie in the overlap of ascending and descending tracks, and often of adjacent tracks, so a given location is imaged several times per 12-day cycle.

Line 48 to 50: These two sentences discusses different issues, variability in the pixel and within the estimation window, and are provided without reference.

The two sentences are now one paragraph that introduces multi-looking, the constant-ΔSWE assumption it makes, and the coherence loss and bias that within-window variability produces (lines 46–52). We have removed the Massonnet and Feigl

(1998) gradient sentence from it. After adding the references suggested above, Rott et al. (2003), Yueh et al. (2026), Touzi et al. (1999), and Zebker and Chen (2005), we felt it added little, and cut it to keep the focus on those studies.

Line 50: Not completely sure if the original quote refers to your problem or phase unwrapping errors, as differences larger than 2pi cannot be recovered with some unwrapping tools.

The quoted limit was the coherence one, derived in Massonnet and Feigl (1998, section 1.3.2, "Gradient") from the necessary condition for interferometry rather than from unwrapping, but as noted in the reply above we have removed the sentence from the manuscript.

Line 74: PDF require definition?

It was the only use of the abbreviation, so we have replaced it with "phase distribution" (line 105).

Line 86: I would suppose that all terms in this section are already noise-free, then how is δ decorrelation noise alone?

They are, and the sentence was wrong. As described in our reply to the Section 2 comment above, the paragraph after Eq. (4) is rewritten so that δ is the noise-free ΔSWE deviation only and decorrelation noise is kept in γ_{other} (lines 102–103).

Line 102: This is repeated.

Removed. The unit-amplitude assumption is stated where Eq. (3) is introduced and its consequences are discussed in the Limitations, so the repetition after the Figure 2 discussion is gone (line 95).

Line 103: not necessarily true for all frequencies.

Agreed. The sentence now restricts the independence, and the multiplicative form of Eq. (5), to L-band and longer wavelengths, and says why (lines 116–120).

Line 164: Int.phase is defined between (-pi, pi].

Corrected to $(-\pi, \pi]$ (line 203). The one other occurrence, in Section 3.5, already used the half-open interval.

Line 328 and 333: Again the effect of the LIA could be affecting the non-stationarity.

The incidence angle can affect the NISAR windows, unlike the lidar analysis where it is held fixed, but its sign is not determined. The cell phase is $\kappa(\theta)\,\Delta$SWE, so within-window variation of the local incidence angle adds to the spread where deeper accumulation coincides with larger κ and partly cancels it where the two are anticorrelated, as the Limitations paragraph states (lines 517–522). The no-accumulation pair shows that terrain geometry alone does not produce the residual. Which way the incidence-angle term acted over the December storm cannot be resolved without a lidar pair coincident with the NISAR dates, which we list as the clearest next step (Section 5.5).

Line 342: I don't agree with this, see my points regarding Section 2.

We have softened this. The sentence now says those areas may not be recoverable at the window sizes of current spaceborne products, rather than at any window size, and grounds it in the window-size sweep (line 408).

Line 406: I think those paper start my [by] calculating coherence spatially, so I don't think saying they use temporal rather than spatial averaging is right, can you check?

Correct, and the sentence was wrong. We were using the main text's averaging language to draw the connection and went too far. Phase linking still forms the coherence spatially but can use smaller spatial windows because of the redundancy of the stack's pairs, which the sentence now says (lines 476–477).

Line 412: drop the formula of the phase closure as neither it or z are defined or used elsewhere.

Dropped. The sentence now describes the closure phase in words, as the sum of the interferometric phases around a triplet of acquisitions (line 480).

185 *Figure 7 refers to sub-pixel on the title.*

Corrected. The figure title now reads “WY2025 (17 days): M by band at 81 m” (now Figure 10), and “sub-pixel” does not appear anywhere in the manuscript.